HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and deployed a **Python**-based agent using **LangChain** to interact with an **LLM API**, enabling it to retrieve and summarize information from external knowledge bases via a **RAG** pattern.
- Engineered a **FastAPI** service in **Python** to expose a fine-tuned **transformer model** for inference, handling approximately 50 requests per second with a p95 latency under 200ms.
- Built an automated data ingestion pipeline using **Python** and **Pandas** to process and clean unstructured text data from diverse sources, preparing it for **LLM** training and analysis.
- Designed and implemented RESTful APIs using **Java Spring Boot** for internal microservices, facilitating data exchange and integration with upstream and downstream systems.
- Contributed to a **Java** application by implementing new features and resolving defects, ensuring code quality through **JUnit** tests and peer code reviews.
- Containerized the AI inference service using **Docker** and deployed it to a **Kubernetes** cluster on **AWS EKS**, configuring basic resource limits and health checks for improved stability.
- Wrote production runbooks and internal documentation for the AI services team, standardizing incident response procedures and reducing average resolution time for P1 incidents by 15%.

SKILLS

AI/ML – LLMs, Generative AI, RAG, LangChain, Prompt Engineering, Transformer Models, Model Inference, Fine-tuning

Programming – Python, Java, SQL, Bash Scripting

Web Frameworks – FastAPI, Flask, Spring Boot, REST APIs

Data Handling – Pandas, NumPy, Data Warehousing, ETL, Data Modelling

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Kubernetes, Jenkins, Git

Databases – PostgreSQL, MySQL, Oracle

BI & Analytics – Power BI, Tableau, DAX, R

PROJECTS

RAG-based Knowledge Retrieval Agent

- Developed a **Python** agent using **LangChain** and **OpenAI API** to implement a Retrieval-Augmented Generation (RAG) system, allowing users to query internal documentation.
- Integrated a vector database (**Pinecone**) to store and retrieve relevant document chunks, improving the accuracy of LLM responses by approximately 30% for factual queries.
- Packaged the agent as a **FastAPI** service, enabling simple integration into existing enterprise applications for enhanced information retrieval capabilities.

AI Model Deployment Pipeline

- Engineered a CI/CD pipeline using **Jenkins** and **Docker** to automate the build, test, and deployment of **Python**-based ML models to a **Kubernetes** environment.
- Implemented model serving using **FastAPI**, ensuring low latency inference for a classification model trained on customer interaction data, achieving ~0.88 AUC on validation sets.
- Configured automated scaling and monitoring for the inference service on **AWS EKS**, maintaining high availability and performance under varying load conditions.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)